

Written Report 2

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Section 3 - Dimensionality reduction techniques

A.Principal Component Analysis (PCA)

Objective

The objective of this analysis is to apply the Principal Component Analysis (PCA) technique to the Online News Popularity dataset in order to reduce its dimensionality by eliminating redundancy among variables related to article content, sentiment, and keyword popularity. Therefore, the resulting principal components will serve as an optimized set of predictors that retain as much of the original variance as possible while simplifying the data structure.

The analysis was focused on 16 variables specifically related to content structure, keyword impact and sentimental tone of the article. We deliberately excluded categorical variables (such as 'data_channel' or 'day_of_week') and LDA topic variables to avoid mixing different types of information.

Procedure

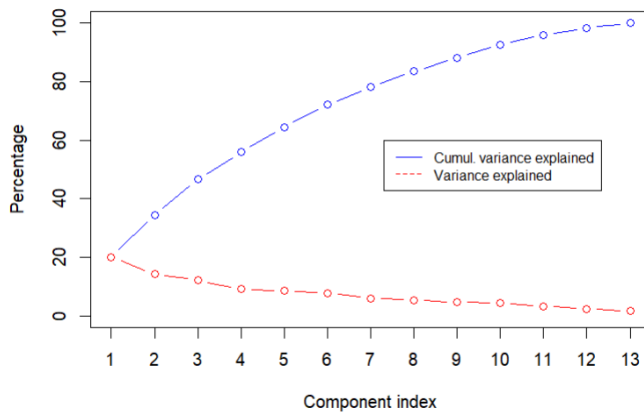
Firstly, before deriving the principal components, we performed a correlation study on the candidate variables with the objective of identifying and eliminating redundancy. Using a correlation matrix, we identified pairs of variables that surpassed the 0.75 cut-off value we set. For each pair of highly correlated variables, we retained only the variable with the strongest relationship to the target variable, "log_shares". Although PCA is an unsupervised learning technique, which means that the target variable is not used, we used this criteria as a variable selection strategy to ensure the variables used in PCA are relevant.

To derive the principal components we first omitted all observations with missing variables (NA values) using the **na.omit()** function. This elimination is crucial as complete data is required to build the matrix standardised observations and the subsequent covariance matrix, which are the initial steps of the eigenvalue decomposition.

Then, we create the matrix of standardised observation using the **scale()** function. This step is critical for calculating the principal components, as it ensures that all variables contribute equally to the analysis regardless of their original scale. Once the data was standardised, we computed the correlation matrix and computed its eigenvalue decomposition using the **eigen()** function. After this, we computed the PC loadings, which are the eigenvectors of the matrix S, previously built. After obtaining the eigenvectors (PC loading), we computed the principal components by performing the matrix multiplication between the standardised matrix of observations and the matrix containing the PC loadings.

Results

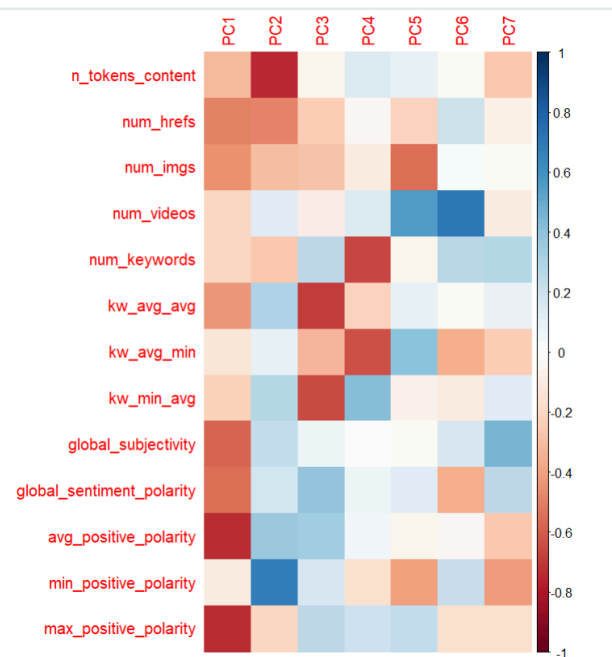
To evaluate the result obtained from the PCA, we calculated the variance and the cumulative explained by each of the principal components. To visualize the results of these statistics, we designed a scree plot, which compares the variance and cumulative variance explained by each component.



From the scree plot, we clearly see that none of the principal components retain more than 20% of variance which means that there is not a dominant variable that retains a vast majority of variance. Based on the 80% criterion, we selected the first 7 components as they approximately retain 79% of the total variance. This indicates that we can represent the original 13 variables in just 7 new variables without a significant loss of information, which accounts for a 46% dimensionality reduction.

Then, we calculated the scaled PC loadings that give us information about the correlation between the original variables and the principal components. This calculation is very helpful to identify which variable drives each of the principal components. We reinforce the analysis with a correlation plot.

- **PC1** (20.14% of variance explained) → The variables with highest correlation are avg_positive_polarity (-0.739), max_positive_polarity (-0.735) and global_subjectivity (-0.577). All correlation values are negative, which indicates that articles with high scores on PC1 tend to have less positive and less subjective content.
- **PC2** (14.35% of variance explained) → The variables with highest correlation are n_tokens_content (-0.741) and min_positive_polarity (0.674). This principal component is a contrast between the article length and the minimum positive polarity.
- **PC3** (12.17% of variance explained) → The variables with highest correlation are kw_avg_avg (-0.686) and kw_min_avg (-0.657). This third component clearly captures information about the minimum average keyword popularity. As in the first component, the most relevant correlations are negative, meaning that articles with high values on PC3 tend to have lower values on those two variables.
- **PC4** (9.21% of variance explained) → The variables with highest correlation are num_keywords (-0.661) and kw_avg_min (-0.637). This component captures how many keywords the article has and the average of the minimum shares of its keywords.
- **PC5** (8.51% of variance explained) → The variables with highest correlation are num_videos (0.557) and num_imgs (-0.558). This component contrasts articles with videos against articles with images.
- **PC6** (7.69% of variance explained) → This component is dominated by the variable num_videos (0.716). This component shows information about articles with a high number of videos.

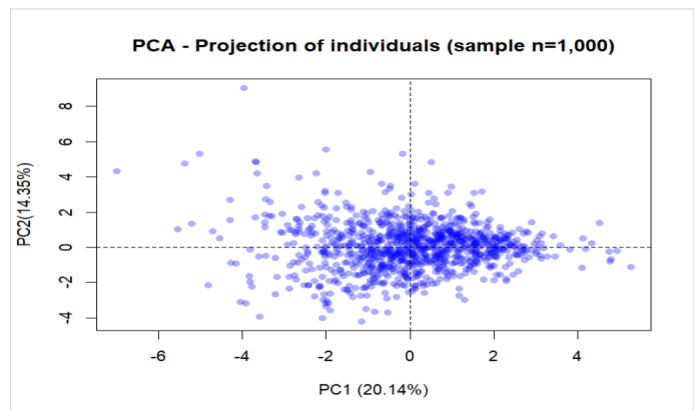


- **PC7** (6.06% of variance explained) → This component is dominated by global_subjectivity (0.460). This component captures subjectivity not explained by previous components.

After the correlation analysis between principal components and original variables, we can conclude that the majority of the variance (**~56% of variance captured by PC1-PC4**) is driven by sentiment-related variables and keyword popularity. This result suggests that keyword strategy and emotional tone of the article are the most informative dimensions of the dataset.

On the other hand , the last components (**PC5 to PC7**) are associated with multimedia content (num_images and num_videos), which collaboratively explain a small percentage of the total variance (**~20%**). This reinforces the idea that multimedia features contribute relatively little to the overall structure of the data.

Following the analysis of the PC and their correlation with original variables, we proceeded to plot the individuals in PC1 and PC2. This visualization allows us to identify potential clusters, outliers, and the overall distribution of observations based on their PC scores. We selected a sample of 1000 individuals due to the dataset size, as the plotting of all observations would overcrowd the plot and it would be difficult to make conclusions.

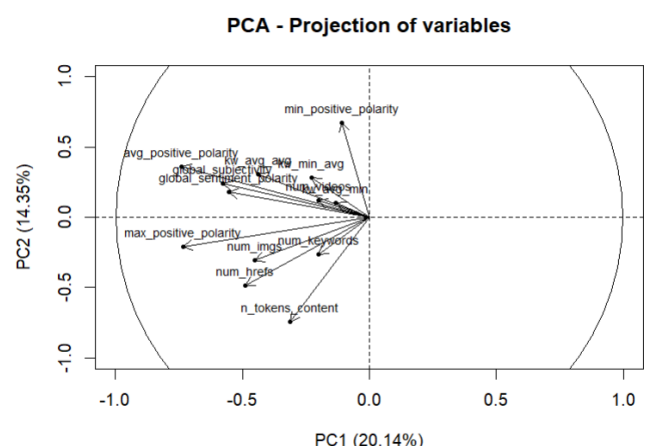


The projection of individuals shows that the majority of the individuals are scattered around zero, which is an indication of the majority of articles having average values on article length, positive polarity and subjectivity. Individuals are more scattered around PC1 which is consistent with the percentage of variance explained, as PC1 explains more variance than PC2. Regarding outliers, some of them are visible both in PC1 and PC2. However, particularly extreme outliers are visible along PC2 which suggests the existence of articles with extreme values of article length and minimum positive polarity.

To reinforce the results above, we used summary statistics to show the distribution of all individuals and not only of the sample ones. In both dimensions we see extreme values which correspond to these outliers identified previously in the visualization.

After this, we plotted the variables in a factor map of variables. The factor map shows the projection of the 13 variables onto PC1 and PC2. Variables which are closer to the origin (represented by the discontinuous lines) are poorly represented by the principal component while variables closer to the edge of the circle (far away from origin) are better represented.

Along PC1, All arrows point to the left meaning that PC1 is negatively correlated with all variables. The longest arrows along PC1 belong to avg_positive_polarity, max_positive_polarity and global_subjectivity, consistent with the correlation analysis performed earlier. On the other hand, along PC2, we have arrows pointing up and down.



This means that PC2 is positively correlated with some variables and negatively with other ones. Talking about arrows, the longest ones are `min_positive_polarity` pointing upward and `n_tokens_content` and `num_hrefs` pointing downward, confirming that PC2 captures the opposition between article length and minimum positive polarity (as said before in the correlation analysis).

Conclusion

The PCA has successfully reduced the dimensionality of the candidate variables from 13 variables to 7 (**46% reduction**) while maintaining a significant percentage of variance explained, which ensures that information is not lost. Apart from this, the analysis reveals that the data structure is divided in three main pillars:

- **Emotional Tone and subjectivity** → The first principal component, the one with the highest percentage of variance explained is strongly correlated with variables regarding global subjectivity and polarity.
- **Article Composition vs. Sentiment** → This is made evident in PC2. There is a clear trade-off between article length (`n_tokens_content`) and minimum positive polarity. This indicates that longer articles may struggle to maintain a consistently positive tone throughout the text.
- **Keyword Strategy** → The weight of this is visible in PC3 and PC4. These components isolate the impact of keywords. They distinguish between the historical success of keywords (`kw_avg_avg`) and the volume/variety of keywords used (`num_keywords`).
- **Extra : Limited weight of multimedia content** → The PCA reveals that multimedia features (images and videos) account for a marginal portion of the total variance.

In conclusion, the PCA analysis reveals that the Online News Dataset is a complex dataset where there are not any individual variables that dominate the explained variability.

B. Correspondence Analysis

Objective

The objective of this analysis is to study whether a statistically significant relationship exists between the thematic channel of an article (the different categories of it: Lifestyle, Entertainment, Business, Social Media, Tech and World) and the day of the week that it was published. Both variables are categorical, making Correspondence Analysis the most appropriate technique, as it allows to visually represent association between qualitative categories and low-dimensional space.

Procedure

Channel information was originally stored as six binary columns and was converted into a single categorical variable using `max.col()`, following the same logic applied to 'day_of_week'. Articles with no channel assigned were removed using `rowSums()`.

We then built the contingency table using `table()` and assessed statistical significance via a chi-square test before proceeding. Row and column profiles were computed to compare category distributions on equal footing. Finally, **CA()** from *FactoMineR* was applied to the contingency table, and the results were visualized with `fviz_ca_biplot()` from *factoextra*.

Results

Chi-square test

The chi-square test returned $X^2 = 44.73$ with 30 degrees of freedom $((7-1) \times (6-1))$ and a p-value of 0.041. Since $p < 0.05$ we reject H_0 and conclude that publication day and the thematic channel are not independent, justifying the CA. The most notable deviations from the expected table appear on weekends, as Lifestyle articles are over-represented on

Saturdays and Sundays, while Business and World articles are under-represented. Weekdays show much smaller deviations, suggesting a more uniform content mix from Monday to Sunday.

Row and Columns profiles

Row profiles show a very similar channel distribution across weekdays (~25 Tech, 21 Entertainment, 17 Business, 19 World, 10 Lifestyle, 7 Social Media), meaning the five working days are almost interchangeable. Weekend days stand out, as Lifestyle reaches ~16% on Saturday and ~15% on Sunday (versus the ~10% on weekdays), while Business drops to ~14% (versus the ~17% on weekdays). Column profiles confirm that Saturday and Sunday carry a proportionally higher weight in Lifestyle than in Business or World.

Eigenvalues and inertia

The CA produces a maximum of 5 dimensions ($\min(7-1, 6-1) = 5$). Dimension 1 explains 97.4% of the total inertia and Dimension 2 a further 2.2%, so the first two dimensions together capture 99.5% of the total association. The two-dimensional factor map is therefore an almost perfect representation of the data.

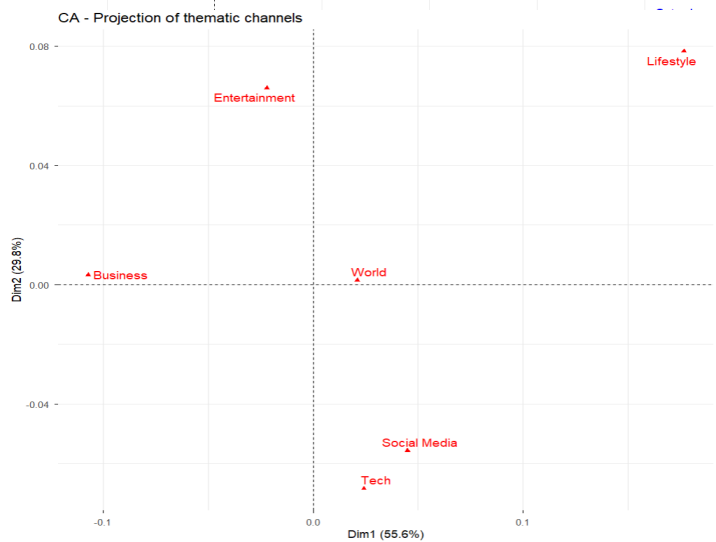
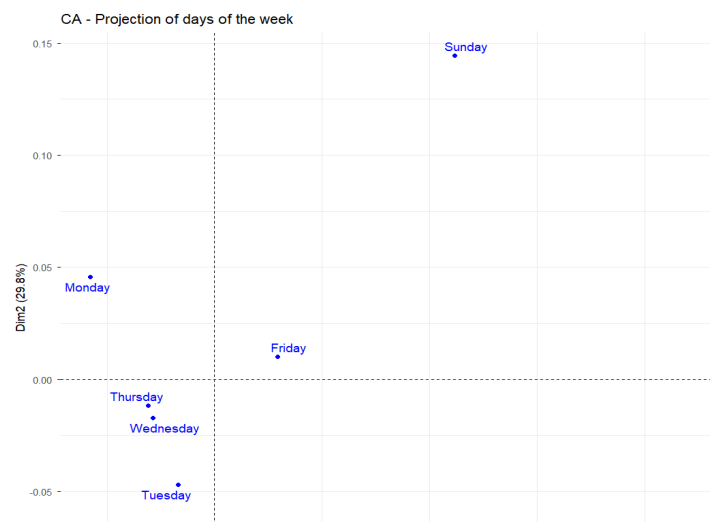
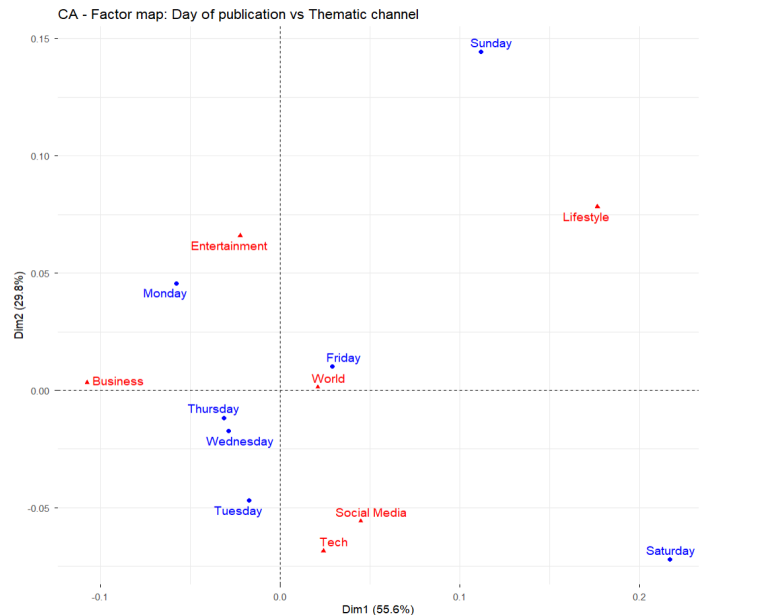
Factor map interpretation

Dimension 1 (97.4%) is almost entirely driven by Saturday (contribution of 59.2%) and Sunday (31.1%) which together make 90.3% of its variation. Both days have large positive coordinates (Saturday +0.201, Sunday +0.162). Lifestyle has the largest channel contribution (62.7%, coordinate +0.137), confirming its strong weekend association. Business (-0.048) and World (-0.044) have negative coordinates, placing them on the weekday side. All five working days cluster near the origin, reflecting their similar and average channel mix.

Dimension 2 (2.2%) is dominated by Monday (contribution of 54.2%) and Entertainment (54.9%), both with negative coordinates, suggesting a mild tendency for Entertainment articles to be over-represented on Mondays compared to the rest of weekdays.

Quality of representation ($\cos^2$)

The $\cos^2$ values confirm that most categories are very well represented in two



dimensions. Saturday (0.998) and Sunday (0.988) are almost perfectly captured by Dimension 1, as are Lifestyle (0.999), Tech (0.938), Business (0.941) and Social Media (0.979). The only exception is Entertainment ($\cos^2$ Dim1: 0.059, Dim2: 0.923) which is instead best captured by Dimension 2, consistent with its uniform distribution across all days.

Conclusion

The CA reveals clear patterns between publication day and thematic channel:

- **Weekend vs weekday divide:** The dominant structure is the contrast between weekend and weekdays, capturing 97.4% of total inertia.
- **Lifestyle is the weekend channel:** Lifestyle articles are distinctly over-represented on Saturdays and Sundays.
- **Business and World are weekday channels:** Both have negative Dimension 1 coordinates, placing them on the opposite side from the weekend days.
- **Weekdays are nearly interchangeable:** The five working days cluster near the origin, with the only exception being a mild Monday-Entertainment association in Dimension 2.
- **Entertainment is uniformly distributed:** It shows no strong day preference and is primarily captured by Dimension 2.

In conclusion, the CA demonstrates that the choice of publication day is not random with respect to the thematic content. Publishers favour lifestyle-oriented content for weekend audiences and shift towards business and world news during the working week, reflecting different reading habits across the week.

Section 4 - Clustering

Clustering Over PCA Results

Objective

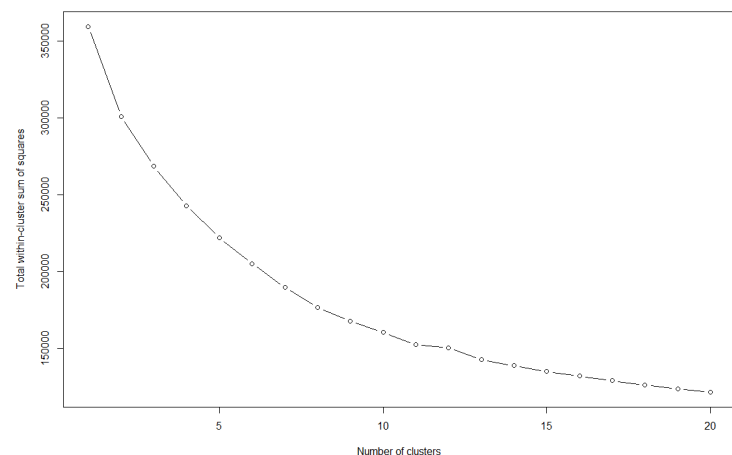
The goal is to identify natural groupings of articles by applying clustering to the principal components obtained in the previous section. Working with the Principal Components allows us to work with a reduced quantity of data while preserving the original structure and reducing computational costs and noise.

K-means clustering:

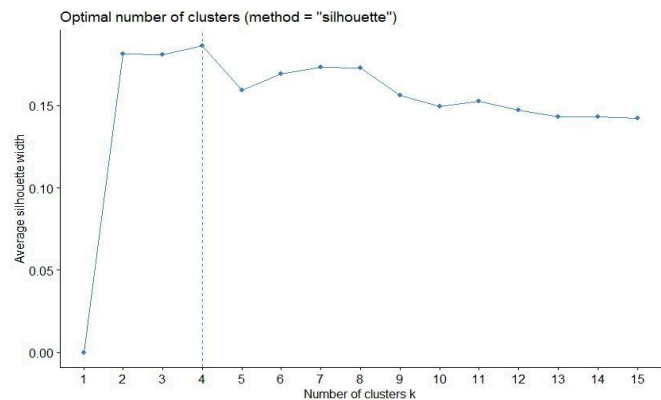
Cluster Selection

The first step in k-means clustering is to select the number of clusters we want to have. To do so, there are different methods that evaluate how good the selected number of clusters are through different criteria. We will use the elbow method and the silhouette method.

The elbow method computes the total Within-cluster Sum of Squares (WSS) of clusters from 1 to n, we have decided to check from one to 20 clusters. The elbow plot shows a smooth and continuous decrease in WSS with no sharp inflection point, which is common in datasets driven by human behaviour. Despite the absence of a clear elbow, the rate of decrease visibly slows around $k=4$ and $k=10$, suggesting these as candidate values.



The silhouette method measures how similar is a given object to the elements of its own cluster compared to other clusters. The graph confirms that $k=4$ is the optimal number of clusters, as it produces the highest average silhouette width of around 0.18. However, this value falls below the 0.25 threshold typically associated with reasonable cluster structure. This makes the analysis weak, but does not invalidate it and makes us decide to select 4 clusters.

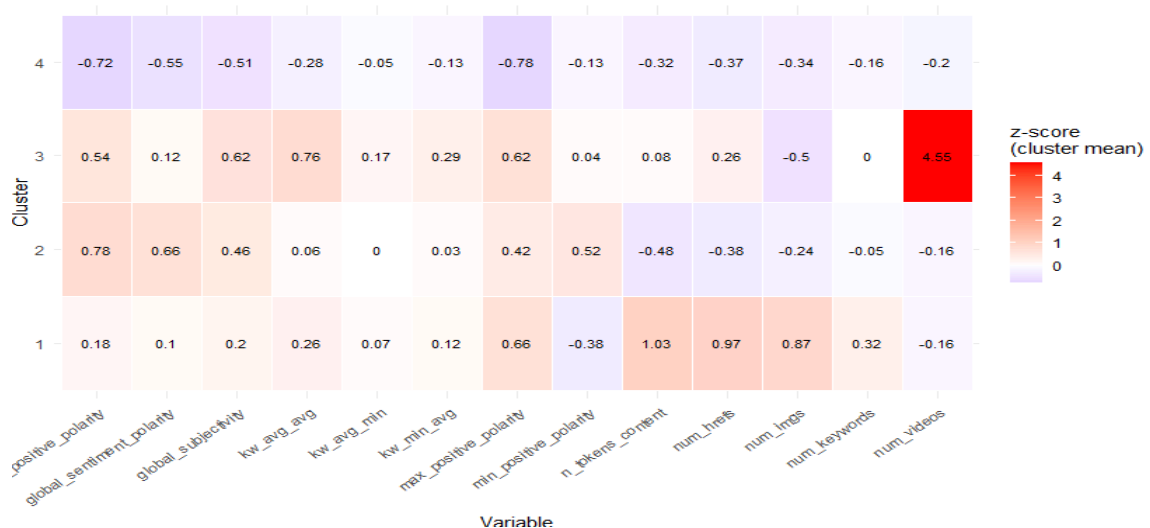


Cluster Interpretation

Once we have decided that 4 clusters is the best, we apply k-means clustering to our compressed data. To interpret the clusters, we examined the mean value of each PC at the cluster centroid, combining the cluster means table with the profile plot, which displays the centroid of each cluster across all 7 principal components and we have reached the following conclusions:

- **Cluster 1** is the smallest cluster, with 1,318 articles. It groups long articles ($n_tokens_content = 1.03$) with many links ($num_hrefs = 0.97$) and many images ($num_imgs = 0.87$). These articles show a high variability in their emotional tone, with a high maximum positive polarity (0.66) but a low minimum positive polarity (-0.38), suggesting that the positivity is not consistent throughout the article.
- **Cluster 2** has 9,183 articles. The articles in this cluster are short ($n_tokens_content = -0.48$), have a very positive tone ($avg_positive_polarity = 0.78$), a high global sentiment polarity ($global_sentiment_polarity = 0.66$) and high subjectivity ($global_subjectivity = 0.46$). Compared to Cluster 1, the positivity is uniform throughout the article, as both the minimum and maximum positive polarity are above average ($max_positive_polarity = 0.42$ and $min_positive_polarity = 0.52$).
- **Cluster 3** has 10,385 articles. This cluster stands out mainly because of the use of a very high number of videos ($num_videos = 4.55$), combined with popular keywords ($kw_avg_avg = 0.76$). Images are below average ($num_imgs = -0.50$), suggesting that videos replace images as the main multimedia format. The tone is somewhat positive and subjective, but that is not the defining characteristic of this cluster.
- **Cluster 4** is the biggest cluster, with 14515 articles. It is characterized by an objectively negative or neutral tone, all polarity and subjectivity variables are consistently below average, with $max_positive_polarity = -0.78$ standing out in particular. These are short articles ($n_tokens_content = -0.32$) with few links ($num_hrefs = -0.37$), few images ($num_imgs = -0.34$), and less popular keywords ($kw_avg_avg = -0.28$). It is the most grey cluster, with no particularly distinctive features above the average.

Cluster Interpretation: Standardised Mean per Variable



Hierarchical clustering

Because the matrix contained 35 400 observations making it impossible to compute by the hierarchical clustering algorithm due to its time and space complexity. To solve this a 2000 article reproducible sample was produced. After filtering extreme outliers that could distort the algorithm's interpretations, the final clean sample consists of 1987 articles.

Distance Metric Selection

Three distance metrics were evaluated: Manhattan, Euclidean and Chebyshev. A Silhouette analysis revealed that Manhattan distance peaked at $k = 2$. Nevertheless, this would split the data into one massive group and one small group, offering limited interpretive value.

Euclidean distance had the largest value for $k = 4$ (this is the number of clusters that will be chosen), providing a more balanced partition. Therefore, Euclidean is the chosen distance metric.

Linkage Method Selection

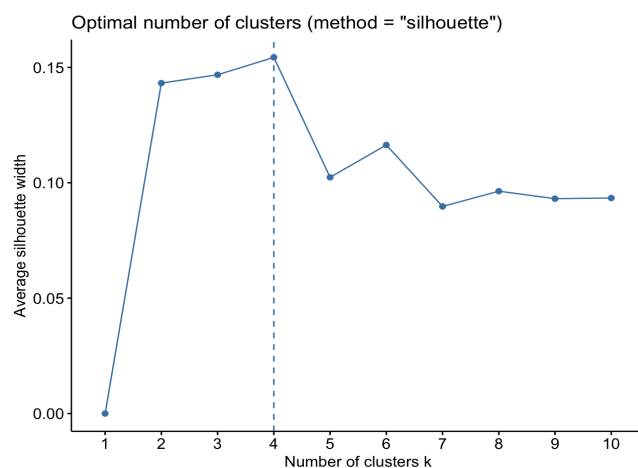
Single and Complete linkage methods were first tested. Both produced highly skewed solutions, placing over 85% of the observations into a single cluster (out of 4 clusters). Single linkage method failed due to chaining: a path of incrementally similar articles forces different articles into one cluster. Complete linkage is theoretically more robust, however the dataset has extreme values. This concludes in clusters that are similar and should be merged together not merged, and the clusters that are really different merged.

Given that the both linkage methods did not produce the results we were looking for, we made additional research to identify a suitable alternative.

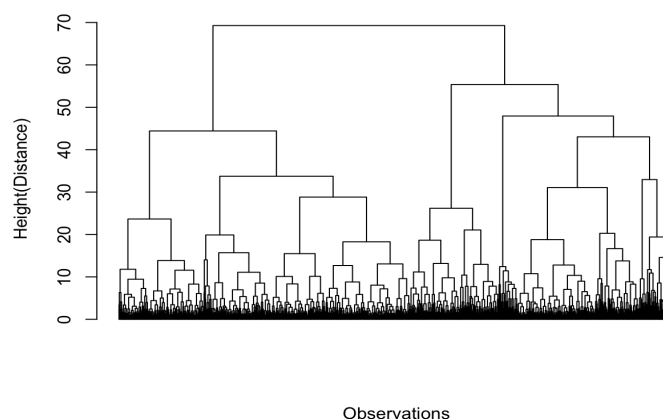
Ward's minimum variance method was identified as an optimal solution. Because Ward's method merges two clusters only when the merge produces the smallest possible increase in total within cluster variance, it is efficient with chaining and outliers. This approach resulted in a more balanced distribution.

Optimal Number of Clusters

To determine the optimal number of clusters, we used two complementary criteria. The Dendrogram analysis showed a notable height gap between dividing the data in 3 or 4 clusters. The Silhouette analysis confirmed that $k = 4$ was the optimal solution, being the highest average silhouette width (0.15). Although an average silhouette width of 0.15 indicates a weak structure, the consistency across both criteria supports our initial thought that $k = 4$ was the best available segmentation.



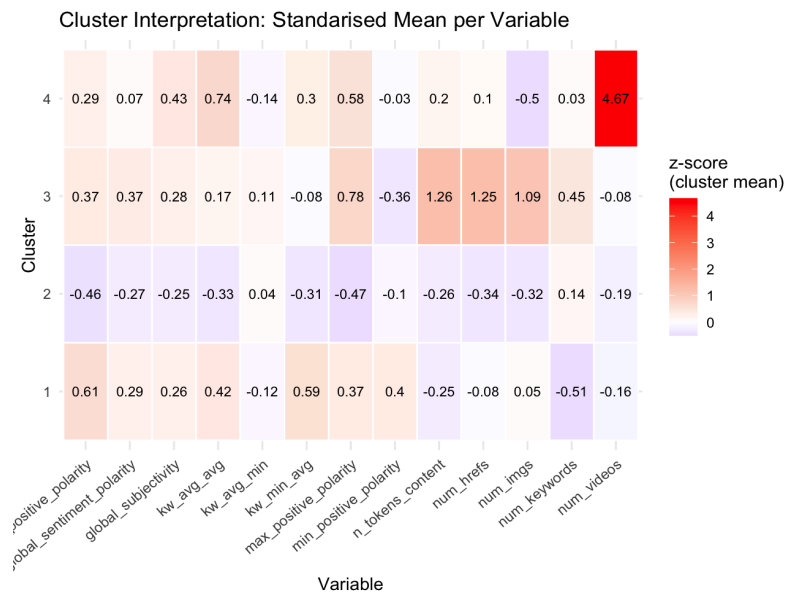
Dendrogram



Cluster Interpretation

To interpret the clusters, we computed the standardised mean values of the variables that formed the PCs for each of the clusters, and visualised the results in the heatmap. Out of the interpretation of the heatmap, we have been able to draw the following conclusions:

- **Cluster 1:** Slightly Emotionally Positive and Poor Keyword Articles. This cluster shows slightly above average polarity and subjectivity. Keyword metrics fluctuate weakly around the mean. In conclusion, the first cluster represents articles that tend to be emotionally flat or slightly positive, and poorly promoted through keywords.
- **Cluster 2:** Objective and Short Articles. These articles show below average polarity and subjectivity. Furthermore, they are shorter than the average article. We can conclude this by the negative mean in the `n_tokens_content`, `num_hrefs` and `num_imgs` variables.
- **Cluster 3:** Long Articles. These articles have the highest means in `n_tokens_content`, `num_hrefs` and `num_images`.
- **Cluster 4:** Video Rich Articles. This cluster maintains average metrics across most variables except for `num_videos`, which is outstandingly high. This represents a subgroup of 67 articles which feature significantly more video content relative to the dataset average.



Conclusion

The Hierarchical clustering model using Euclidean distance, Ward's linkage and $k = 4$ reveals the segmentation of the articles into four groups. Although the average silhouette width indicates that the boundaries between clusters are weak, the clusters still capture meaningful patterns.

Method comparison

Both methods independently selected $k = 4$ as the optimal number of clusters. This means that the groups we found are probably correct because we got the same result with both methods. Both also produced a low average silhouette width, 0.18 for K-means and 0.15 for Hierarchical, consistently indicating that cluster boundaries are diffuse, which reflects the continuous nature of editorial profiles in online news rather than a failure of either method.

Although K-means was applied to the full dataset and Hierarchical clustering only to a sample of 2000 articles, both methods identified four very similar clusters, suggesting that the sample is sufficiently representative of the full dataset.

In conclusion, the consistency of results across two independent methods with different algorithmic approaches strengthens the validity of the identified cluster structure.

K-Means Clustering Over CA Results

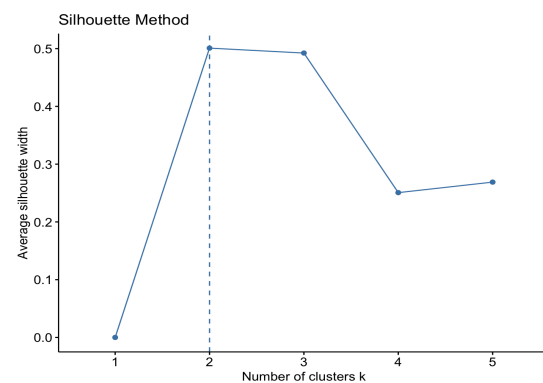
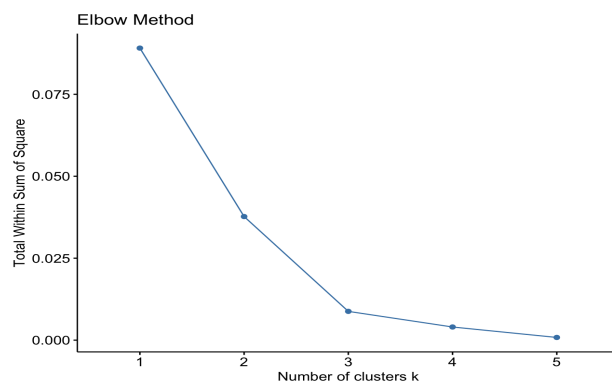
Objective

The goal is to determine which days share a similar editorial profile and to verify if the weekday weekend divide is the primary structural trend in the dataset.

To achieve our goal, we use the data we obtained from the CA analysis on section 3. The input for our clustering model is the coordinates of the two main dimensions of the CA. Retaining the two dimensions is justified because Dimension 1 explains 97.4% of the total inertia and Dimension 2 the 2.2%.

K-Means Clustering Model

The number of clusters has been decided through the analysis of the elbow and the silhouette method. The silhouette method weakly recommended $k = 2$, while the elbow method suggested $k = 3$. We ultimately selected $k = 2$. The justification for this decision is based on the inertia explained by the CA dimensions. Selecting $k = 3$ forces a separation between Saturday and Sunday. However, this separation between them lies along Dimension 2, which explains only 2.2% of the total inertia.



Conclusion

As it was foreseen, the K-means model partitioned the data into the weekday-weekend division.

The weekday cluster groups days with a channel mix dominated by Business. Instead the weekend cluster groups days associated with Lifestyle content. This clustering output successfully confirms our initial hypothesis: the main editorial division in online news is between weekdays and weekends.

